VENKATESHWARLU SARANGI, Ph.D

Hong Kong • LinkedIn • Github • venky.sarangi@gmail.com • CityU HK

PROFESSIONAL SUMMARY

AI/ML Engineer with PhD in Physics and 4+ years of industry experience in developing distributed AI platforms, specializing in end-to-end AI lifecycle from data processing to deployment and monitoring. Expert in translating business problems into measurable AI tasks, establishing baselines and KPIs such as Precision/Recall/F1, MAE, latency, and QPS for key use cases in NLP, Deep Learning, RAG systems, and Large Language Models. Proficient in Python, PyTorch, TensorFlow, scikit-learn, databases, and web frameworks, with bonus experience in LangChain, Docker/K8s, and regulatory compliance in AML and fraud detection.

TECHNICAL SKILLS

Machine Learning & AI Expertise: Machine Learning, Deep Learning, Natural Language Processing, Large Language Models, Supervised/Unsupervised Learning, Feature Engineering, Model Selection & Validation, Hyperparameter Optimization, Time Series Forecasting, Anomaly Detection, Statistical Modeling, Scikit-learn, XGBoost, LightGBM.

RAG & Information Retrieval: Dense/Sparse Retrieval, Vector DBs (Chroma, FAISS, Pinecone, Weaviate, Milvus), Reranking & Hybrid Search, Embedding Models, Query Expansion, Document Chunking & Preprocessing, Grounding & Hallucination Detection, Evaluation Metrics, LangChain, LangGraph.

MLOps & Infrastructure: DVC, MLflow, Docker, Kubernetes (K8s), Helm; CI/CD (GitHub Actions, Jenkins, GitLab CI); Terraform, Weights & Biases; Monitoring (Prometheus, Grafana, Evidently); Feature Stores (Feast, Tecton); A/B Testing & Experimentation; Blue-Green & Canary Deployments; Model Registry & Versioning.

Cloud Platforms: AWS (SageMaker,IAM, EC2, EKS, S3, Lambda), Azure, Machine Learning, GCP Vertex AI; Serverless & Edge Deployment.

Development & Tooling: Python, PyTorch, NumPy, Pandas; FastAPI, Flask; Git, GitHub; Streamlit, Gradio (Prototyping); SQL, Spark, Airflow (Data Pipelines).

WORK EXPERIENCE

DATA SCIENTIST & ML ENGINEER — *Freelancer at XRadio Ltd, HK*

(Aug, 2025 – March, 2026)

Deep Learning and Multimodal Systems for ECG Signal Intelligence

- Built an explainable deep learning model for 12-lead **ECG arrhythmia classification,** addressing challenges from non-stationary signals and inter-patient variability.
- Evaluated multiple **deep learning and vision architectures** including ResNet34, ResNet50, DenseNet, and VGG16, **Vision Transformers (ViT) and Vision-Language Models (VLMs),** multimodal AI systems for healthcare prediction tasks.
- Achieved strong model performance with ResNet34, reaching **AUROC 0.98 and F1-score 0.826** across nine arrhythmia categories.

DATA SCIENTIST & RESEARCH ENGINEER — *Gense Technologies Ltd, HK*

(Oct, 2022 – Aug, 2025)

Projects & Technical Experience

- Designed and deployed end-to-end supervised learning pipelines (Random Forest, XGBoost, Gradient Boosting, logistic regression) for clinical prediction tasks, achieving 87% sensitivity and >99% Laspecificity on imbalanced biomedical datasets.
- Engineered feature extraction and signal processing workflows using Python (NumPy, Pandas, Scikit-learn) to improve SNR and diagnostic quality in time-series bio-conductivity data.
- Achieved a 100% improvement in skin electrode contact impedance without conductive gels, improving signal sensitivity.
- Developed and validated signal enhancement and impedance analysis models for ECG and EIT datasets, improving signal-to-noise ratio and diagnostic reliability.
- Built automated MLOps workflows with Docker, CI/CD, and MLflow for experiment tracking, model versioning, and reproducible deployments across cloud environments. Applied statistical validation frameworks (k-fold cross-validation, ROC-AUC optimization, hypothesis testing) to ensure diagnostic-grade model reliability and regulatory compliance.
- Developed synthetic data augmentation strategies (SMOTE, bootstrapping) to address limited clinical datasets, improving model generalization by 20%. Proficient in Python, SQL, PyTorch, Scikit-learn, and cloud ML platforms (GCP, Firebase, AWS SageMaker) for scalable production systems.

DATA SCIENTIST AND COMPUTATIONAL ANALYST — *Anvipro IT Solutions, India*

(December, 2021 – June, 2022)

Materials Science & Computational Modeling

- Conducted physics-based simulation and data analysis using **COMSOL Multiphysics** to study material flexibility (>90%), stress distribution, hardness, and structural behavior.
- Performed materials **data preprocessing,** cleaning, and simulations using MATLAB and Python basic libraries to improve analysis quality and model readiness.
- Applied **supervised learning and statistical modeling** techniques to analyze material properties, identify trends, and support predictive insights.
- Improved simulation and analytical workflows through **iterative model refinement, results evaluation,** and reliability analysis for engineering and materials-focused projects.
- **Technologies:** Python, Data processing, Feature Engineering, EDA,NumPy, Pandas, Scikit-learn, OriginLab, SQL, Power BI,Statistical Modeling.

PH.D RESEARCH SCHOLAR

CityU HK

(August, 2015 – October 2021)

Materials Engineering / Computational Modeling / Data Analysis

- Designed piezoelectric electrodes, applied advanced statistical and computational modeling for material structure analysis using X-ray and neutron diffraction datasets.
- Performed **Rietveld refinement and data processing, feature engineering, and regression-based parameter optimization, fine-tuning methods, hyperparameter tuning, model optimization** to extract structural insights from complex datasets.
- Published research in **Nature Communications, Communications Physics, Physical Review B, Scientific Reports**, and other high-impact journals (Google Scholar).
- **Technologies:** FullProf and GSAS-II for structural analysis (GUIs) and Pair Distribution Function (PDF) analysis, Python, MATLAB, OriginLab, Scikit-learn, Statistical Modeling.

JOURNAL PUBLICATIONS

- S. Venkateshwarlu et al., **Nature Communications Physics**, (2020).
- S. Venkateshwarlu et al., **Journal of Materials Research**, Cambridge University Press JMR-0830 (2019).
- S. Nayak, S. Venkateshwarlu, et al., **Journal of the American Ceramic Society**, (2019).
- Frederick, Sarangi Venkateshwarlu, et al., **Chemistry of Materials** (2021).
- A. Pramanick, S. Venkateshwarlu, et al., **Physical Review B**, (2021).
- Liang, Zhuoxin Liu, S. Venkateshwarlu, et al., **Nature**, **Light: Science & Applications**, (2021).
- G. Srinivas, S. Venkateshwarlu, et al., **Applied Physics Letters**, (2015).
- S. Venkateshwarlu, et al., **Journal of Modern Materials**, (2016).
- Pramanick, Venkateshwarlu, **Journal of the European Ceramic Society**, (2023).
- Nirmal and Sarangi Venkateshwarlu et al., **arXiv:2508.10940**, (2025).
- Kwok, W.C., Sarangi Venkateshwarlu et al., **Nature Scientific Reports** (2025).

EDUCATION | TRAINING - CERTIFICATIONS

IIT Bombay India and CityU HK,

- **Ph.D.** Materials Science and Eng (Physics), City University of Hong Kong, (Oct, 2021)
- **M.Tech.** Materials Science and Eng (Physics), IIT Bombay, India, (June, 2013)
- **Generative AI with Large Language Models (LLMs)**
- DeepLearning.AI & Amazon Web Services | Coursera | February 2026
- Skills: Transformers Architecture (FLAN-T5, BERT, GPT, DeepSeek, Qwen, Phi-2, Gemma) · RL models · PyTorch · PEFT · Fine-tuning · LoRA · QLoRa · Knowledge Distillation(KD) · RAG · LangChain · LangGraph · Deployment · vLLM · OpenWebUI · FastAPI· Streamlit · RunPOD· LightningAI · HuggingFace,· Optuna Optimization
- **Machine Learning Specialization** | August 2025
- **Deep Learning and NLP Specialization** | January 2026
- Stanford University & DeepLearning.AI | Coursera
- **End-to-End MLOPS Bootcamp** | Udemy | April 2026-ongoing